

Food Bank Surplus Redistribution Platform

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SECTION: 5E

1. Functional & Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow Donor Partners to post surplus food packages with quantity, food type, expiry time, and dietary tags.	High	Pass: A donor can successfully create and publish a food listing with all required details. Fail: The listing is published without required information.	Allows donors to make surplus edible food available to verified shelters before expiry.
FR-002	Functional	The system shall track the expiry time of each posted food batch and automatically deactivate batches after expiry.	High	Pass: An expired food batch is removed from the active claim board. Fail: An expired batch remains available for reservation.	Prevents shelters from reserving food that is no longer within its valid distribution period.
FR-003	Functional	The system shall notify eligible nearby shelters when a relevant surplus food batch is posted.	High	Pass: Verified shelters within the applicable notification area receive a notification. Fail: Eligible shelters do not receive the notification.	Helps redistribute perishable food quickly and reduces food wastage.
FR-004	Functional	The system shall allow Shelter Coordinators to reserve an available food batch and select a pickup window.	High	Pass: A shelter can reserve an available batch and receive a confirmed pickup window. Fail: The system allows multiple shelters to reserve the same batch.	Ensures that surplus food is allocated to a shelter in an organized manner.
FR-005	Functional	The system shall generate and validate a pickup verification code when a reserved food batch is collected.	High	Pass: A valid verification code confirms pickup and updates the reservation status. Fail: An invalid code confirms the pickup.	Provides a reliable way to confirm that the reserved food was actually collected.
NFR-001	Non-Functional	The platform shall dispatch notifications to shelters within a 5 km radius promptly when high-quantity perishable food is listed.	High	Pass: Benchmarking confirms that notifications are delivered within the defined response target under simulated peak load. Fail: Notifications exceed the target response time.	Rapid notification is important because perishable food has a limited usable period.
NFR-002	Non-Functional	The platform shall maintain reliable access to active food listings and reservation information during periods of high usage.	High	Pass: Simulated peak-load testing confirms that listings and reservations remain accessible without loss of confirmed reservation data. Fail: Active reservations or listings are lost or become inaccessible during peak usage.	Reliability is necessary to prevent failed pickups and loss of food redistribution opportunities.

2. Use-Case Flow Specification

Use Case ID	UC-001	Primary Actor	Shelter Coordinator
Use Case	Reserve Food Batch	Goal	Reserve an available surplus food batch and select a suitable pickup window before the food expires.

2.1 Preconditions

1. The Shelter Coordinator is registered and verified on the platform.
2. At least one valid food batch is available for reservation.
3. The selected food batch has not expired.
4. The Shelter Coordinator has access to the platform.

2.2 Postconditions

1. The selected food batch is reserved for the Shelter Coordinator.
2. The selected pickup window is recorded.
3. The reserved batch is no longer available for other shelters.
4. The reservation confirmation is provided to the Shelter Coordinator.

2.3 Main Success Scenario

1. The Shelter Coordinator logs into the platform.
2. The system displays available surplus food batches.
3. The Shelter Coordinator selects an available food batch.
4. The system displays the food batch details, including quantity, dietary information, and expiry time.
5. The Shelter Coordinator chooses the desired food batch.
6. The system displays available pickup windows.
7. The Shelter Coordinator selects a pickup window.
8. The system validates that the food batch is still available and has not expired.
9. The system confirms the reservation.
10. The system marks the food batch as reserved and records the selected pickup window.
11. The system displays the reservation confirmation to the Shelter Coordinator.

2.4 Alternate Flow — A1. Food batch expires before reservation is completed

1. The Shelter Coordinator attempts to reserve the selected food batch.
2. The system detects that the food batch has expired.
3. The system prevents the reservation.
4. The system removes the expired food batch from the active claim board.
5. The system asks the Shelter Coordinator to select another available food batch.

3. Food Bank Use Diagram

